

1. Online Hitting Set for Intervals

(a) Sophia: $\Omega(n)$ lower bound

Let $S = \{1, 2, \dots, n\}$. Adversary reveals:

$$A_1 = [1, n], \quad A_2 = [1, n-1], \quad \dots, \quad A_n = [1, 1]$$

Sophia's trace: A_1 unhit $\implies$ add $\max(S \cap A_1) = n$. $A_2 = [1, n-1]$ contains no added point $\implies$ add $n-1$. Continuing: Sophia adds $n, n-1, \dots, 1$. Total: $|H_{\text{Sophia}}| = n$.

OPT: $1 \in A_i$ for all $i \implies H^* = \{1\}$, $|H^*| = 1$.

$$\boxed{\text{ALG/OPT} = n/1 = \Omega(n)}$$

(b) Tammy: $O(\log n)$ upper bound

Let $H^* = \{p_1^*, \dots, p_k^*\}$ be OPT, $|H^*| = k$. Tammy adds midpoint $m = \lceil (\min(S \cap A) + \max(S \cap A))/2 \rceil$ when A unhit.

Charging: When Tammy adds point m for unhit interval A , charge m to any $p^* \in H^* \cap A$ (exists since H^* hits A).

Window invariant: For each $p^* \in H^*$, let W_{p^*} = set of points in S between the nearest added points flanking p^* (initially $W_{p^*} = S$). Suppose new interval $A \ni p^*$ is unhit. Then A contains no added point $\implies A \cap S \subseteq W_{p^*}$. Tammy adds midpoint m of $A \cap S$. Since $m \notin A$ (not yet added when A processed) wait... m being added NOW means:

Tammy adds m . After this, m is closer to p^* than any previously-added point flanking p^* , so window shrinks:

$$|W_{p^*}^{\text{new}}| \leq \frac{1}{2}|A \cap S| \leq \frac{1}{2}|W_{p^*}^{\text{old}}|$$

Bound: $|W_{p^*}| \leq n$ initially. After $\lceil \log_2 n \rceil$ charges, $|W_{p^*}| = 0 \implies$ no future interval containing p^* can be unhit. So each p^* absorbs $\leq \lceil \log_2 n \rceil$ charges.

$$|H_{\text{Tammy}}| \leq k \lceil \log_2 n \rceil \implies \boxed{\text{ALG/OPT} = O(\log n)}$$

2. Online Unit Covering

(i) Leftmost: unbounded

Adversary: $\varepsilon < 1/n$; points arrive as $x_i = -(i-1)\varepsilon$ for $i = 1, \dots, n$.

Leftmost's trace: $x_1 = 0$: add $[0, 1]$. $x_2 = -\varepsilon$: uncovered $\implies$ add $[-\varepsilon, 1 - \varepsilon]$. Continuing: each x_i uncovered by previous intervals (all extend rightward from $x_j, j < i$) $\implies$ add $[x_i, x_i + 1]$. Total: n intervals.

OPT: All points in $[-(n-1)\varepsilon, 0]$, span < 1 . Single interval $[-(n-1)\varepsilon, 1 - (n-1)\varepsilon]$ covers all.

$$\boxed{\text{ALG/OPT} = n/1 \rightarrow \infty \text{ as } n \rightarrow \infty}$$

(ii) Centered: 2-competitive

Let $c_1, c_2, \dots, c_a$ be centers of ALG intervals in arrival order.

Claim 1: $\forall j < k: |c_k - c_j| > 1/2$.

c_k uncovered when arrived, but $[c_j - 1/2, c_j + 1/2] \in C$ at that time $\implies c_k \notin [c_j - 1/2, c_j + 1/2] \implies |c_k - c_j| > 1/2$.

Claim 2: Every unit interval I contains ≤ 2 centers.

Suppose $c_p < c_q < c_r$ in I (length ≤ 1). Then $c_r - c_p \leq 1$. But $c_q - c_p > 1/2$ and $c_r - c_q > 1/2 \implies c_r - c_p > 1$. Contradiction.

Competitive ratio: Every ALG center lies in some OPT interval, and each OPT interval contains ≤ 2 centers.

$$\boxed{\text{ALG} \leq 2 \cdot \text{OPT}}$$

(iii) Grid: 2-competitive

Grid cell $k = [k, k + 1)$ for $k \in \mathbb{Z}$. ALG opens ≤ 1 interval per cell visited.

Claim: Each OPT interval $[a, a + 1]$ intersects ≤ 2 grid cells.

$[a, a + 1]$ has length 1, spanning at most 2 consecutive grid cells $\lfloor a \rfloor$ and $\lfloor a \rfloor + 1$.

Competitive ratio: Each ALG cell contains a point covered by some OPT interval. Each OPT interval accounts for ≤ 2 ALG cells.

$$\boxed{\text{ALG} \leq 2 \cdot \text{OPT}}$$

3. Online Unit Clustering

Algorithm (Grid)

When x_i arrives: let $k = \lfloor x_i \rfloor$. If a cluster labeled k exists, add x_i to it. Else create new cluster labeled k containing $\{x_i\}$.

Validity: Every cluster labeled k contains points in $[k, k + 1) \implies \text{diameter} < 1 \implies \text{coverable by unit interval } [k, k + 1]$. $\checkmark$

Competitive analysis

Let OPT partition points into k^* clusters $C_1^*, \dots, C_{k^*}^*$, each coverable by a unit interval.

Claim: Each C_j^* contributes points to ≤ 2 ALG clusters.

$C_j^* \subseteq [a_j, a_j + 1] \implies [x] \in \{\lfloor a_j \rfloor, \lfloor a_j \rfloor + 1\} \forall x \in C_j^*$. So points of C_j^* land in ≤ 2 grid labels.

Bound: ALG clusters = distinct grid labels used. Each OPT cluster contributes ≤ 2 labels:

$$|\text{ALG}| \leq \sum_{j=1}^{k^*} 2 = 2k^* = 2 \cdot |\text{OPT}|$$

$$\boxed{\text{ALG}/\text{OPT} \leq 2}$$